

MULTICORP FIRE PROTECTION SERVICES

TechForce Pro Application Blueprint

A workforce management platform purpose-built for Multicorp — turning shop days into billable revenue and giving supervisors real-time visibility across every tech, truck, and ticket.

FOR INTERNAL REVIEW — DRAFT V2.0

This document is ready for your review. Please mark up any additions, removals, or changes you'd like — once approved, we move straight to production.

Prepared by: Development Team

Date: May 6, 2026

Confidential — Multicorp Internal Use Only

INTERNAL MEMORANDUM

To: Management — Multicorp Fire Protection Services
From: Development Team
Date: May 6, 2026
Re: TechForce Pro — Application Blueprint for Review

Per your direction, I've put together the complete application blueprint for **TechForce Pro** — our custom workforce management platform built specifically for Multicorp's operations.

This document covers everything we discussed: the scheduling system, employee management, job tracking, the AI-powered features, and — most importantly — the **shop day management system** that balances technician needs with company profitability. I know shop days are a key concern for the techs, so I made sure that section is thorough.

Here's what I need from you:

- **Review each section** and let me know what resonates and what needs to change
- **Add anything I missed** — features, rules, edge cases you've seen in the field
- **Remove anything that doesn't fit** our current priorities or budget
- **Flag the must-haves vs. nice-to-haves** so we can prioritize the sprint plan

Once you sign off on the blueprint (with your changes), we begin production immediately. The 16-week sprint timeline at the back of this document gets us to a working system by late August 2026.

I'm confident this platform will make a real difference — especially in converting shop days to billable work and giving you full visibility into tech profitability. Looking forward to your feedback.

Contents

FOUNDATION

01	Company Profile & Business Context	5
02	System Architecture & Platform Overview	7
03	Dashboard & KPI Command Center	9

CORE OPERATIONS

04	Shop Day Management & Profitability Engine	12
05	Scheduling & Auto-Assignment	16
06	Employee Management & Tech Profitability	19
07	Job Management — Statuses, Returns & Reschedules	22

REVENUE & AI FEATURES

08	AI Scheduler & Smart Assignment	25
09	Service Ticket Scanning & Invoice Generation	28
10	Customer Pricing & Contract Management	31
11	Reporting & Analytics Dashboards	33

DELIVERY

12	Tech Stack & Security	36
13	16-Week Sprint Timeline	38
14	Next Steps & Sign-Off	41

APPENDIX

A	Shop Day Profitability Calculator (Sample)	43
B	Service Code Reference (H, 6, R, I, C, N)	44

Company Profile & Business Context

Understanding Multicorp's operations, team structure, and the daily challenges TechForce Pro is designed to solve.

About Multicorp Fire Protection Services

Multicorp Fire Protection Services is a full-service fire protection company headquartered at **9693 Gerwig Lane, Suite O, Columbia, MD 21046**. The company provides inspection, testing, maintenance, and installation services for fire suppression systems, fire extinguishers, sprinkler systems, exit lighting, and related life-safety equipment.

The company serves a diverse client base including government facilities (Baltimore County), commercial properties, restaurants, schools, healthcare facilities, and residential group homes across Maryland.

Current Team Structure

TECHNICIAN	ROLE	CERTIFICATIONS	BILL RATE	SALARY
Sarah Johnson	Suppression Lead	Suppression, Extinguisher	\$1,600/day	\$85K
Derek Williams	Sprinkler Tech	Sprinkler	\$1,200/day	\$70K
Marcus Taylor	Extinguisher Tech	Extinguisher	\$900/day	\$55K
Kevin Park	Extinguisher Tech	Extinguisher, Exit Lights	\$900/day	\$55K
Angela Davis	Suppression Tech	Suppression	\$1,400/day	\$75K
James Rodriguez	Helper / Apprentice	Extinguisher	\$500/day	\$40K

The Problem We're Solving

Right now, Multicorp manages scheduling, job tracking, and invoicing through a mix of paper tickets, spreadsheets, and manual processes. This creates several pain points:

Operational Gaps

- No real-time visibility into who's where and what's billable
- Shop days accumulate without a system to convert them to revenue
- Handwritten tickets require manual data entry for invoicing
- Returns and reschedules get lost in the shuffle

Revenue Leaks

- Techs on shop days when open jobs need their cert
- No profitability tracking per technician
- Customer pricing inconsistencies across service calls
- Delayed invoicing from paper-to-digital lag

The Goal: TechForce Pro replaces all of this with one platform — giving you a clear picture of every tech, every job, and every dollar, while automatically turning idle shop days into billable work.

System Architecture & Platform Overview

How the platform is built, what each layer does, and why these choices matter for Multicorp's scale.

Architecture at a Glance

TechForce Pro is a modern web application accessible from any browser — desktop, tablet, or phone. No software to install. Updates deploy instantly.

LAYER	TECHNOLOGY	WHAT IT DOES
Frontend	React + TypeScript	The interface supervisors and techs interact with daily
Backend	Convex (real-time database)	Stores all data, handles business logic, syncs instantly across devices
AI Engine	GPT-4 Vision + Custom Models	Reads handwritten tickets, auto-generates invoices, smart scheduling
Hosting	Vercel (cloud)	Fast, reliable, scales automatically — 99.99% uptime
Auth	Role-based access	Supervisors see everything; techs see their own schedule and jobs

Key Design Principles

Real-Time Everything

When a job status changes in the field, the dashboard updates instantly. No refresh needed. Supervisors always see the current state — not what it was 10 minutes ago.

Mobile-First Field Access

Techs access their daily schedule, update job statuses, and scan tickets from their phone. The interface is designed for one-handed use in the truck.

Offline Capable

Ticket scans and job updates queue locally when signal drops (basements, parking garages) and sync when connectivity returns. No lost data.

Built for Multicorp's Scale

The system handles 6 techs today and scales to 60+ without architecture changes. Adding a new tech is a 30-second form, not a database migration.

User Roles & Permissions

ROLE	ACCESS LEVEL	KEY CAPABILITIES
Owner / GM	Full access	All features, pricing, employee management, reports, system settings
Supervisor	Operations	Schedule, jobs, dispatch, employee performance, shop day management
Technician	Personal	Own schedule, job updates, ticket scanning, training status
Office Staff	Admin	Invoicing, customer records, reporting, data export

REVIEW NOTE

Are there other roles you'd like in the system? For example, a read-only "Client Portal" where customers like BCOGov can see their service history and upcoming inspections?

Dashboard & KPI Command Center

The first screen you see every morning — real-time numbers that tell you exactly how the business is running.

Top-Line KPIs

Four headline metrics sit at the top of the dashboard. Each updates in real time as jobs complete and statuses change.



Profit Leak Alerts

The dashboard proactively flags money left on the table. These aren't buried in reports — they're front and center, first thing you see:

- **Marcus Taylor has 4/5 shop days used this month** — one more and he's over budget. The system suggests reassigning open extinguisher jobs to him.
- **James Rodriguez at 10/12 shop days** — his utilization is 78%. The system flags open jobs he's certified for and recommends auto-assigning them.
- **Derek Williams has 2 shop days next week** — there's an open sprinkler job at Metro Office Park that matches his cert. One-click assign.

HOW THIS SAVES MONEY

Every shop day costs the company a tech's daily salary with zero revenue. A \$55K tech sitting in the shop costs ~\$212/day. If the system converts just 2 shop days per tech per month to billable work, that's **\$2,500+/month in recovered revenue** across the team.

Today's Dispatch Board

A live view of every job dispatched today — who's assigned, what time, current status (Pending → In Progress → Completed / Return / Reschedule). Supervisors can drag-and-drop reassignments if a tech calls out.

Weekly Schedule Snapshot

Color-coded grid showing the full week at a glance. Green = billable day, red = shop day, yellow = training. You can spot patterns instantly — if a tech has 3 shop days in a row, something needs attention.

Employee Profit Table

Every tech listed with their revenue YTD, net profit, utilization %, and shop day count. Sortable by any column. The supervisor can instantly see who's carrying the team and who needs more jobs.

Shop Day Management & Profitability Engine

The heart of TechForce Pro — making sure every tech gets the shop days they need while keeping the company profitable.

Why This Section Matters: Shop days are important to techs — they need time to maintain equipment, restock trucks, handle paperwork, and train. But every shop day is a day with zero revenue. TechForce Pro balances both sides: techs get their allocated shop days, and the company maximizes billable utilization.

Monthly Shop Day Allocation

Each technician receives a set number of shop days per month based on their role, seniority, and certification level. This allocation is configurable by management and visible to both the tech and the supervisor.

TECH	ROLE / CERT LEVEL	MONTHLY ALLOCATION	DAILY COST	MONTHLY SHOP COST	TARGET UTILIZATION
Sarah Johnson	Suppression Lead	2 days/month	\$327	\$654	90%+
Derek Williams	Sprinkler Tech	3 days/month	\$269	\$808	85%+
Marcus Taylor	Extinguisher Tech	3 days/month	\$212	\$635	85%+
Kevin Park	Extinguisher Tech	3 days/month	\$212	\$635	85%+
Angela Davis	Suppression Tech	2 days/month	\$288	\$577	90%+
James Rodriguez	Helper / Apprentice	4 days/month	\$154	\$615	80%+

REVIEW NOTE

These allocations are starting recommendations based on role and cert level. You can adjust any tech's monthly allocation at any time. The system will recalculate profitability targets automatically.

How Shop Day Profitability Works

The system calculates a **break-even threshold** for each tech — the minimum number of billable days needed per month to cover their salary and shop day cost.

The Math (Example: Marcus Taylor)

METRIC	VALUE	HOW IT'S CALCULATED
Annual Salary	\$55,000	Set by management
Monthly Salary Cost	\$4,583	$\$55,000 \div 12$
Working Days/Month	~22	Standard
Shop Day Allocation	3 days	Set by management
Available Billable Days	19 days	$22 - 3 \text{ shop days}$
Bill Rate	\$900/day	Based on cert level
Max Monthly Revenue	\$17,100	$19 \times \$900$
Monthly Profit (at 100% util.)	\$12,517	$\$17,100 - \$4,583$
Break-Even Days	5.1 days	$\$4,583 \div \900

Marcus needs just **5.1 billable days per month** to cover his salary. Everything above that is profit. With 19 available billable days, there's significant margin — but every unnecessary shop day erodes it.

Shop Day Tracking Dashboard

Each tech's profile shows a visual progress bar for shop days used vs. allocated this month:

TECHNICIAN	USED / ALLOCATED	STATUS	ACTION
Sarah Johnson	1 / 2	On Track	—
Derek Williams	2 / 3	On Track	—
Marcus Taylor	4 / 3	Over Limit	Auto-assign open extinguisher jobs
Kevin Park	3 / 3	At Limit	Monitor — no more shop days this month
Angela Davis	1 / 2	On Track	—
James Rodriguez	10 / 4	Critical	Immediate: assign all available jobs

Automatic Shop Day → Billable Conversion

This is TechForce Pro's most powerful feature. When a tech is scheduled for a shop day but there are open jobs matching their certification, the system automatically suggests:

1. **Checks open job board** — are there unassigned jobs needing this tech's cert?
2. **Checks customer priority** — HIGH priority jobs get offered first
3. **Checks geography** — nearest jobs preferred to minimize drive time
4. **Sends notification** — supervisor gets an alert: "Marcus has a shop day Tuesday but Patapsco HS needs an extinguisher round. Assign?"
5. **One-click assign** — supervisor approves, schedule updates, customer gets notified

The Result: No escort or call-ahead needed. Open jobs auto-assign to techs with excess shop days. The system does the thinking — the supervisor just approves or overrides.

Shop Day Rules Engine

Management sets the rules. The system enforces them automatically:

RULE	DESCRIPTION	CONFIGURABLE?
Monthly Cap	Max shop days per tech per month	Yes — per tech or per role
Consecutive Limit	Max consecutive shop days before a billable day is required	Yes — default: 2
Week Balance	No more than X shop days per week	Yes — default: 1
Auto-Convert Threshold	When shop days exceed X% of allocation, auto-suggest conversions	Yes — default: 75%
Training Exemption	Training days don't count toward shop day limits	Yes — on/off
Override Authority	Who can approve extra shop days beyond allocation	Supervisor / Owner

REVIEW NOTE

Which of these rules matter most to you? Are there additional rules you'd want? For example: "Apprentices cannot have a shop day on Monday" or "Lead techs get priority for choosing their shop day." Please mark up what you'd like.

Financial Impact of Shop Day Management

Here's what disciplined shop day management means for the bottom line:

Current State (Without TechForce Pro)

METRIC	CURRENT	MONTHLY COST
Avg shop days per tech	~5/month (estimated)	—
Total team shop days	~30/month	—
Total shop day cost	—	\$7,200/month
Lost revenue opportunity	~30 × avg \$950 bill rate	\$28,500/month potential

With TechForce Pro (Projected)

METRIC	TARGET	MONTHLY IMPACT
Avg shop days per tech	~3/month (managed)	—
Total team shop days	~17/month	—
Shop days converted to billable	~13/month	+\$12,350 revenue
Shop day cost reduction	43% fewer shop days	-\$3,100 saved
Net monthly impact		+\$15,450/month

Annual projected impact: +\$185,000 in recovered revenue and reduced shop day costs. This alone pays for the entire platform many times over.

What Techs See

Transparency matters. Techs can see their own shop day dashboard:

- How many shop days they've used this month vs. their allocation
- Their utilization percentage and how it compares to their target
- Upcoming scheduled shop days
- Why a shop day was converted (if it was) — "Reassigned to Chipotle Rt 1 — Hood Suppression Check, HIGH priority"

This keeps techs informed and removes the feeling of shop days being taken away arbitrarily. The system is fair, transparent, and consistent.

REVIEW NOTE

Should techs be able to **request** specific shop days (e.g., "I need a shop day Friday to restock my truck")? If so, should it go through supervisor approval? Let me know and I'll add this to the workflow.

Scheduling & Auto-Assignment

The weekly schedule is the backbone of operations — who goes where, when, and why.

Weekly Schedule View

The schedule shows a full Monday–Friday grid with every technician as a row. Each cell is color-coded:

- Green** — Billable day (assigned to a job, showing revenue amount)
- Red** — Shop day (in the shop, zero revenue)
- Yellow** — Training day (certification, safety, apprentice training)

Supervisors can click any cell to see the full job details, reassign the tech, or convert a shop day to a billable assignment.

Week Total Revenue

Each row shows the tech's weekly revenue total. The bottom of the schedule shows the **Week Total Revenue** — the combined billable output of the entire team. For the sample week of June 9–13, 2026, the total is **\$28,600**.

Open Jobs — Available for Assignment

Below the schedule sits the **Open Jobs** board. These are jobs that need a tech but haven't been assigned yet. Each shows:

FIELD	DESCRIPTION
Priority	HIGH (red), MED (orange), LOW (gray)
Job Description	What needs to be done
Client	Who requested the service
Cert Needed	Which certification the tech must hold
Status	Unassigned (ready to fill)

Open jobs auto-assign to techs with excess shop days — no escort or call-ahead needed. The supervisor gets a notification and can approve or override.

Schedule Rules

- Techs can only be assigned jobs matching their certification
- No tech gets double-booked (one job per day unless explicitly overridden)
- HIGH priority open jobs surface first in auto-assignment suggestions
- The system warns if assigning a job would push a tech over 5 consecutive billable days (fatigue prevention)

REVIEW NOTE

Should supervisors be able to schedule **half-day** assignments (e.g., billable in the morning, shop in the afternoon)? Some techs may need to split the day. Let me know if this is a real scenario for Multicorp.

Employee Management & Tech Profitability

Every tech is a profit center. This module tracks the numbers that matter.

Employee Overview

The Employees page opens with four summary stats across the entire team:



Individual Tech Cards

Each technician has a detailed card showing:

FIELD	WHAT IT SHOWS	WHY IT MATTERS
Status	On Job / Available / Shop Day	Real-time location awareness
Certifications	Suppression, Extinguisher, Sprinkler, Exit Lights	Determines which jobs they can take
Revenue YTD	Total billed amount this year	Top-line contribution
Net Profit	Revenue minus salary cost	True profitability per tech
Utilization %	Billable days ÷ total working days	Efficiency metric — target 85%+
Shop Days	Used / Allocated with visual bar	Shop day budget tracking
Salary	Annual compensation	Cost basis for profit calculations
Bill Rate	Daily billable rate	Revenue per billable day

Profitability Ranking

The system automatically ranks techs by net profit contribution. This isn't about pitting techs against each other — it's about knowing where to invest in training, where to add jobs, and where shop days are eating into margins.

#	TECH	REVENUE YTD	NET PROFIT	UTILIZATION	ROI
1	Sarah Johnson	\$140K	\$106K	98%	2.65x
2	Angela Davis	\$126K	\$96K	94%	2.68x
3	Derek Williams	\$108K	\$80K	91%	2.54x
4	Kevin Park	\$86K	\$64K	92%	2.56x
5	Marcus Taylor	\$81K	\$59K	87%	2.47x
6	James Rodriguez	\$41K	\$25K	78%	2.03x

REVIEW NOTE

Should profitability data be visible only to management, or should techs see their own numbers? Some companies find transparency motivating; others prefer to keep financials private. What's right for Multicorp?

Job Management — Statuses, Returns & Reschedules

Every job has a lifecycle. TechForce Pro tracks it from dispatch to completion — including the messy middle when things go wrong.

Job Lifecycle

Every job moves through a clear status flow:

STATUS	MEANING	NEXT STEPS
Pending	Assigned but tech hasn't started	Tech marks "In Progress" on arrival
In Progress	Tech is on-site working	Mark complete, return, or reschedule
Completed	Job done, ticket signed	Scan ticket → generate invoice
Return	Needs a follow-up visit	Schedule return, reason tracked
Reschedule	Couldn't complete — customer issue, weather, etc.	Auto-suggest new date

Returns Workflow

Returns are a reality in fire protection — maybe a part was ordered, maybe the system needs a second visit. TechForce Pro tracks every return:

- Tech marks job as "Return" with a reason code (Parts Needed, Follow-Up Required, Customer Request, Failed Inspection)
- The system automatically creates a follow-up job linked to the original
- Supervisor gets a Returns queue showing all outstanding follow-ups
- Return jobs inherit the original job's priority + 1 level (a MED becomes HIGH)

Reschedule Workflow

When a job can't happen as planned:

- Tech or supervisor marks "Reschedule" with a reason (No Access, Weather, Equipment Failure, Customer Canceled)
- The system suggests the next available date based on the tech's schedule
- Customer is notified automatically (if enabled)
- Reschedule count is tracked — jobs rescheduled 3+ times get flagged for supervisor attention

Job Detail View

Each job record contains everything needed for the service call:

- Client name, address, contact information
- Service type and description
- Assigned tech and certification required
- Priority level and time window
- Revenue amount and billing status
- Full history — every status change, note, and timestamp
- Linked service tickets and invoices

REVIEW NOTE

What are the most common return reasons at Multicorp? I want to make sure the reason codes in the system match what actually happens in the field. Should we add "Permit Required" or "Fire Marshal Hold" as options?

AI Scheduler & Smart Assignment

Let the system do the thinking — auto-assign jobs, optimize routes, and flag scheduling conflicts.

How the AI Scheduler Works

The AI scheduler runs continuously, analyzing the schedule and open job board to recommend optimal assignments. It considers:

Assignment Factors

- **Certification match** — the tech must hold the required cert
- **Availability** — is the tech scheduled for a shop day that can be converted?
- **Shop day balance** — techs over their shop day limit get priority for open jobs
- **Customer priority** — HIGH priority jobs assigned first
- **Geography** — nearest tech to job site preferred

Protection Rules

- **Fatigue prevention** — no tech works more than 5 consecutive billable days without a break
- **Training days** — never converted to billable (protected)
- **Minimum shop days** — each tech is guaranteed a minimum shop day floor (configurable)
- **Override always available** — supervisor can always override any AI suggestion

Auto-Dcline Rules

The system can automatically decline or deprioritize certain assignments based on rules you set:

RULE	EXAMPLE	WHY
Distance limit	No jobs > 45 miles from base	Drive time eats into billable hours
Revenue floor	No jobs under \$400 for suppression techs	High-cert techs should take high-value jobs
Client blacklist	No assignments to [specific client]	Payment history, access issues
Cert mismatch	Never assign exit light jobs to suppression techs	Cert integrity

Emergency Schedule Override

When an emergency call comes in — fire marshal inspection, insurance deadline, critical system failure — the supervisor can trigger an **Emergency Schedule** mode:

- All available techs (including shop day techs) surface instantly
- System shows who's closest, who has the right cert, and who's least impacted by reassignment
- One-click reassignment with automatic notifications
- Original schedule preserved as "rescheduled" — nothing gets lost

REVIEW NOTE

How often do emergency overrides happen at Multicorp? Daily? Weekly? This affects how prominently we feature the Emergency Schedule button. Right now it's a red button in the top-right of the schedule view.

Service Ticket Scanning & Invoice Generation

From handwritten ticket to professional invoice in seconds — no manual data entry.

The Current Process

Today, the workflow looks like this:

1. Tech fills out a paper service ticket in the field (handwritten)
2. Ticket goes back to the office (end of day or end of week)
3. Someone manually reads the handwriting and enters it into the billing system
4. Invoice gets created, reviewed, and sent to the customer
5. Time from service to invoice: **3–7 days (or more)**

The TechForce Pro Process

1. Tech completes the service and fills out the paper ticket (same as today — no behavior change)
2. Tech snaps a photo of the ticket with their phone
3. AI reads the handwriting — extracts customer, line items, quantities, service codes (H, 6, R, I, C), and PO number
4. System cross-references **Customer Pricing** tables to auto-fill unit prices
5. Professional invoice generated instantly — ready to email or export
6. Time from service to invoice: **under 2 minutes**

Service Code Recognition

The AI understands Multicorp's service code system from the ticket columns:

CODE	MEANING	INVOICE LABEL	PRICING IMPACT
H	Hydrotest	Hydrostatic Test	Premium — requires certification & equipment
6	6-Year Maintenance	6-Year Maintenance Service	Higher than standard — full teardown & rebuild
R	Recharge	Extinguisher Recharge	Standard service rate
I	Inspection	Annual / Semi-Annual Inspection	Base inspection rate
C	Condemned	Unit Condemned — Replacement Needed	Triggers replacement quote
N	Non-Compliance	Non-Compliance Notice Issued	Follow-up required — no charge for notice

When multiple codes are circled on the same line (e.g., R + I), the system generates separate line items for each service — or a combined rate if the customer pricing table specifies one.

Proof of Concept

We've already demonstrated this capability using Multicorp Ticket #21154 (Glen Arm Maintenance Facility, BCOGov). The AI successfully extracted all 13 line items, identified circled service codes (R, R+I, 6+R), applied appropriate pricing, and generated a professional invoice — total \$1,025.55 including MD sales tax.

REVIEW NOTE

Does Multicorp use any other service codes beyond H, 6, R, I, C, N? Also — should the invoice auto-email to the customer's billing contact, or should it always go through office review first?

Customer Pricing & Contract Management

Different clients pay different rates. TechForce Pro keeps pricing consistent and auditable.

Per-Client Rate Tables

Every customer in the system has a pricing profile. When a job is assigned or an invoice is generated, the system automatically pulls the correct rate for that client and service type.

CLIENT	SUPPRESSION	EXTINGUISHER	SPRINKLER	EXIT LIGHTS	CONTRACT TYPE
Olive Garden — Columbia	\$1,400	\$600	—	\$400	Annual
Lincoln Elementary (Patapsco)	—	\$900	—	\$500	Semi-Annual
Metro Office Park	\$1,500	\$700	\$1,200	\$450	Annual
Harbor Condos	\$1,200	\$500	\$900	\$600	Annual
BCOGov (Glen Arm)	\$1,400	\$900	\$1,100	\$500	Government
Chipotle — Rt 1	\$1,400	\$600	—	\$350	Annual

Pricing Features

- **Contract-based pricing** — rates locked for the contract period; auto-flag when renewal is due
- **Volume discounts** — configurable by quantity (e.g., extinguisher inspections at scale)
- **Service-specific rates** — different prices for inspection vs. recharge vs. 6-year maintenance
- **Tax handling** — Maryland sales tax auto-applied where applicable; government contracts exemption supported
- **Rate history** — full audit trail of price changes with date and who approved the change

REVIEW NOTE

Should certain clients get a "cost-plus" model (Multicorp's cost + markup %) vs. fixed rates? Also — do government contracts like BCOGov have different invoicing requirements (e.g., specific forms, PO references, net-60 terms)?

Reporting & Analytics Dashboards

The numbers you need to make decisions — not buried in spreadsheets, but live and visual.

Available Reports

REPORT	WHAT IT SHOWS	FREQUENCY
Revenue Summary	Total revenue by tech, client, service type, and time period	Daily / Weekly / Monthly
Utilization Report	Billable vs. shop vs. training days per tech	Weekly / Monthly
Shop Day Analysis	Shop days used vs. allocated, cost impact, conversion opportunities	Weekly / Monthly
Profitability by Tech	Revenue minus salary, ranked by net contribution	Monthly / Quarterly
Profitability by Client	Revenue per client vs. cost to service (tech time, travel)	Monthly / Quarterly
Returns & Reschedules	Frequency, reasons, resolution time, repeat offenders	Weekly
Invoice Aging	Outstanding invoices by age (30/60/90 days), payment trends	Weekly
Certification Tracker	Cert expiry dates, training needed, compliance status	Monthly

Export & Sharing

- All reports exportable as PDF or Excel
- Scheduled email delivery — e.g., weekly utilization report every Monday at 7 AM
- Dashboard sharing — create read-only links for ownership review

Custom Date Ranges

Every report supports flexible date filtering: this week, this month, this quarter, YTD, or any custom date range. Compare period-over-period (this month vs. last month) to spot trends.

REVIEW NOTE

Are there specific reports your boss or the owner looks at regularly? If there's a "Monday morning report" or "end-of-month summary" format they prefer, I can build it in as a one-click export.

Tech Stack & Security

What's under the hood and how your data stays safe.

Technology Stack

COMPONENT	TECHNOLOGY	WHY THIS CHOICE
Frontend	React 18 + TypeScript	Industry standard, fast, maintainable, huge talent pool
UI Framework	Tailwind CSS + shadcn/ui	Clean, professional look with minimal custom CSS
Backend / Database	Convex	Real-time sync, serverless, no database management needed
AI / OCR	GPT-4 Vision API	Best-in-class handwriting recognition and document understanding
Hosting	Vercel	Instant deploys, global CDN, 99.99% uptime SLA
Authentication	Convex Auth (role-based)	Secure login with role-based permissions
PDF Generation	Server-side rendering	Professional invoices and reports generated on demand

Security Measures

- **HTTPS everywhere** — all data encrypted in transit (TLS 1.3)
- **Data at rest** — encrypted database storage
- **Role-based access** — techs see only their data; supervisors see their team; owner sees everything
- **Audit logging** — every action tracked with user, timestamp, and IP
- **Regular backups** — automated daily backups with 30-day retention
- **SOC 2 compliant infrastructure** — Convex and Vercel both maintain SOC 2 Type II compliance

Data Ownership

All data belongs to Multicorp. Full export available at any time in standard formats (CSV, JSON, PDF).
No vendor lock-in — if you ever want to move to a different platform, your data comes with you.

16-Week Sprint Timeline

From approved blueprint to production-ready system in 16 weeks. Here's the plan.

SPRINT	WEEKS	DELIVERABLES	STATUS
Sprint 1	Weeks 1–2	Foundation — Auth, database schema, employee records, base UI shell	Ready to start
Sprint 2	Weeks 3–4	Schedule — Weekly grid, color-coded days, manual job assignment	—
Sprint 3	Weeks 5–6	Shop Day Engine — Allocation rules, tracking dashboard, utilization bars, profitability math	—
Sprint 4	Weeks 7–8	Job Management — Status workflow, returns, reschedules, dispatch board	—
Sprint 5	Weeks 9–10	AI Features — Ticket scanning, auto-invoice, smart assignment suggestions	—
Sprint 6	Weeks 11–12	Customer Pricing — Per-client rate tables, contract management, tax handling	—
Sprint 7	Weeks 13–14	Reporting — All dashboards, PDF/Excel export, scheduled reports	—
Sprint 8	Weeks 15–16	Polish & Launch — Testing, training, data migration, go-live	—

Key Milestones

- **Week 2:** Internal demo — basic employee and schedule views working
- **Week 6:** Shop day management fully operational — biggest ROI feature live for testing
- **Week 10:** Ticket scanning demo with real Multicorp tickets — invoice generation working
- **Week 14:** Full system operational in staging — all features testable
- **Week 16:** Production launch — team trained and using the system daily

Target Launch: If we begin production in mid-May 2026, the system goes live by **late August 2026**. The shop day engine (Sprint 3) delivers ROI starting **Week 6** — before the full system is even complete.

REVIEW NOTE

Should any sprint be moved up in priority? If shop day management is the #1 pain point, we could build that in Sprint 1–2 and have it running within a month. Let me know your priorities.

Next Steps & Sign-Off

What we need from you to move from blueprint to production.

Your Review Checklist

Please review each section and mark your feedback:

SECTION	YOUR CALL	NOTES / CHANGES
01 — Company Profile	☐ Approve ☐ Change	
02 — System Architecture	☐ Approve ☐ Change	
03 — Dashboard & KPIs	☐ Approve ☐ Change	
04 — Shop Day Management	☐ Approve ☐ Change	
05 — Scheduling	☐ Approve ☐ Change	
06 — Employee Management	☐ Approve ☐ Change	
07 — Job Management	☐ Approve ☐ Change	
08 — AI Scheduler	☐ Approve ☐ Change	
09 — Ticket Scanning & Invoicing	☐ Approve ☐ Change	
10 — Customer Pricing	☐ Approve ☐ Change	
11 — Reporting	☐ Approve ☐ Change	
12 — Tech Stack & Security	☐ Approve ☐ Change	
13 — Sprint Timeline	☐ Approve ☐ Change	

What Happens After Your Review

1. **You mark up this document** — additions, removals, changes, questions
2. **We incorporate your feedback** — updated blueprint within 48 hours
3. **Final sign-off** — you approve the revised blueprint
4. **Production begins** — Sprint 1 starts immediately after sign-off
5. **Bi-weekly demos** — you see working software every 2 weeks throughout the build

The system is designed around what Multicorp actually does every day. The shop day management alone should pay for the platform. The AI invoice scanning eliminates hours of manual data entry per week. And the real-time dashboard means you'll never have to ask "where is everyone?" again.

Looking forward to your feedback. Once you're ready, we go straight to production.

APPENDIX A

Shop Day Profitability Calculator

This table shows the financial impact of each shop day for every technician. Use it to validate the allocations in Section 04.

TECH	SALARY	DAILY COST	BILL RATE	SHOP DAYS/ MO	BILLABLE DAYS	MAX REVENUE	NET PROFIT	ROI
Sarah J.	\$85K	\$327	\$1,600	2	20	\$32,000	\$24,917	4.52x
Derek W.	\$70K	\$269	\$1,200	3	19	\$22,800	\$16,967	3.91x
Marcus T.	\$55K	\$212	\$900	3	19	\$17,100	\$12,517	3.73x
Kevin P.	\$55K	\$212	\$900	3	19	\$17,100	\$12,517	3.73x
Angela D.	\$75K	\$288	\$1,400	2	20	\$28,000	\$21,750	4.48x
James R.	\$40K	\$154	\$500	4	18	\$9,000	\$5,667	2.70x

Each additional shop day beyond allocation costs:

TECH	LOST REVENUE (BILL RATE)	DAILY SALARY COST	TOTAL COST OF 1 EXTRA SHOP DAY
Sarah Johnson	\$1,600	\$327	\$1,927
Derek Williams	\$1,200	\$269	\$1,469
Marcus Taylor	\$900	\$212	\$1,112
Kevin Park	\$900	\$212	\$1,112
Angela Davis	\$1,400	\$288	\$1,688
James Rodriguez	\$500	\$154	\$654

Total cost of the entire team taking **1 extra unplanned shop day** in a month: **\$7,962**

Service Code Reference

Standard service codes used on Multicorp field tickets. These codes appear in columns on the service ticket form and are circled by the technician to indicate work performed.

CODE	FULL NAME	DESCRIPTION	PRICING TIER
H	Hydrotest	Hydrostatic pressure test per DOT / NFPA requirements. Required at specific intervals (5yr/12yr depending on cylinder type).	Premium
6	6-Year Maintenance	Complete teardown, internal inspection, and reassembly of stored-pressure extinguishers. Required every 6 years per NFPA 10.	Above Standard
R	Recharge	Refill extinguishing agent after use or after 6-year maintenance. Includes pressure test and seal replacement.	Standard
I	Inspection	Annual or semi-annual visual inspection per NFPA 10. Check pressure, condition, signage, accessibility, tamper seal.	Base
C	Condemned	Unit fails inspection or test — removed from service. Replacement quote generated automatically.	No charge (triggers replacement)
N	Non-Compliance	Unit or area does not meet code. Notice issued to building owner/manager. Follow-up visit required.	No charge (notice only)

Multiple Codes Per Unit

Techs may circle multiple codes for a single unit. Common combinations:

- **R + I** — Recharge and Inspection (most common combination)
- **6 + R** — 6-Year Maintenance plus Recharge (required after 6yr service)
- **H + R** — Hydrotest and Recharge (required after hydro)
- **I only** — Inspection only (unit passed, no service needed)

TechForce Pro reads each circled code and generates the appropriate invoice line items with correct pricing per the customer's rate table.

Sample AI-Generated Invoice

This invoice was generated automatically from a photo of Multicorp Service Ticket #21154. The AI extracted all line items, identified circled service codes, and applied customer pricing.

SOURCE TICKET

Ticket #21154 — Glen Arm Maintenance Facility (BCOGov)
P.O. #24979 · Job #107977G142 · Tech: TM/EM · Date: 02/10/2026 · Service Type: Returns

QTY	DESCRIPTION	SERVICE	UNIT PRICE	AMOUNT
2	20 lb. CO2 Extinguisher	R	\$55.00	\$110.00
1	15 lb. CO2 Extinguisher	R I	\$60.00	\$60.00
6	5 lb. Pressurized ABC Ext.	R	\$16.79	\$100.75
1	10 lb. Pressurized ABC Ext.	6 R	\$38.00	\$38.00
6	20 lb. Pressurized ABC Ext.	R I	\$36.00	\$216.00
3	20 lb. Pressurized ABC Ext.	6 R	\$42.00	\$126.00
3	Conductivity Test — CO2	Test	\$25.00	\$75.00
19	Tamper Seals	Parts	\$1.50	\$28.50
19	Service Collars	Parts	\$2.00	\$38.00
19	O-Rings	Parts	\$1.25	\$23.75
19	Valve Stems	Parts	\$3.50	\$66.50
1	Site Service Charge (SSC)	Service	\$85.00	\$85.00

Subtotal \$967.50
MD Sales Tax (6%) \$58.05
Total Due \$1,025.55

This invoice was generated in under 2 minutes from a phone photo of the handwritten service ticket. No manual data entry was required. The AI correctly identified all circled service codes (R, R+I, 6+R) and applied BCOGov-specific pricing.